

The Jacobian Conjecture after the Three-Dimensional Counterexample: Exact Validation, a General Seed Family, the Geometry of Non-Properness, and Two-Dimensional Equivariant Rigidity

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Abstract

On 2026-07-19, Levent Alpöge announced (crediting the language model Claude Fable 5 and a question posed by Akhil) an explicit polynomial map $F : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ with constant Jacobian determinant -2 and a repeated value, refuting the Jacobian conjecture for every $N \geq 3$. We report the results of an exact-arithmetic experimental program built on this event. First, we validate the counterexample independently and compute the full fiber over the announced target: it consists of exactly the three announced points. Second, we reverse-engineer the structure of F : it is a weighted skew-product over the invariants $v = xy$, $t = x^2z$, its Keller property reduces to a two-variable identity, and its fibers are governed by a cubic in a single composite variable. Third, we derive and verify a general constructor: every admissible triple (seed polynomial p , scale k , section q) yields a Keller map with determinant $-kp(1)^2$ and generic fiber degree $\deg p + 1$; we exhibit new explicit counterexamples with rational collision certificates in fiber degrees 4, 5 and 6, plus a section-tail instance beyond the announced family shape, prove the degree law $(5d - 3, 5d - 4, 4)$ within the family, and show the announced map is its smallest member. Fourth, we make the failure geometry exact: a preimage escapes to infinity precisely when its fiber root is a multiple root, so the asymptotic variety of F is the plane $C = 0$ together with the explicit discriminant surface $27A^2C^2 - 18ABC + 16A + B^3C - B^2 = 0$; over $\mathbb{R}$ this wall separates a three-sheet from a one-sheet region, and F restricts to a surjective, non-injective real Keller map. The counterexamples reduce at suitable primes to non-injective Keller maps over $\mathbb{F}_\ell$ of degree below the characteristic. Fifth, we prove that the mechanism cannot reach the still-open two-variable case: for arbitrary weights, every $\mathbb{G}_m$ -equivariant Keller map of $\mathbb{C}^2$ is linear (a classification with a positive-factor kill at the top degree), so a planar counterexample, if any, must be genuinely non-equivariant. All computations are exact (rational or symbolic) and reproducible from the repository; every claim below is labeled as machine-verified, derived, or conjectural.

1 Introduction

Let k be a field of characteristic 0. A polynomial map $F = (F_1, \dots, F_N) : k^N \rightarrow k^N$ is a *Keller map* if $\det JF$ is a nonzero constant. The Jacobian conjecture $JC(N)$, stated in two variables by

Kraus in 1884 (with a flawed proof whose gap, control of ramification at infinity, remained the subject's central obstacle [6]) and popularized by Keller in 1939 [4], asserted that every Keller map has a polynomial inverse. Over $\mathbb{C}$, invertibility is equivalent to injectivity, so a refutation requires exactly one repeated value. The classical positive ladder includes Wang's degree-2 theorem, Moh's verification of $JC(2)$ up to degree 100, the Bass–Connell–Wright reduction to cubic homogeneous form [2], and Drużkowski's cubic-linear form [3]; Pinchuk [5] refuted the strong real variant by a non-proper local diffeomorphism with non-constant Jacobian. The conjecture is Problem 16 of Smale's list [7].

On 2026-07-19/20, Alpåge announced the map, with $u = 1 + xy$,

$$F = (u^3z + y^2u(4 + 3xy), \quad y + 3xu^2z + 3xy^2(4 + 3xy), \quad 2x - 3x^2y - x^3z), \quad (1)$$

with $\det JF \equiv -2$ and $F(0, 0, -\frac{1}{4}) = F(1, -\frac{3}{2}, \frac{13}{2}) = F(-1, \frac{3}{2}, \frac{13}{2}) = (-\frac{1}{4}, 0, 0)$ [1]. Appending identity coordinates refutes $JC(N)$ for all $N \geq 3$; the case $N = 2$ remains open. At the time of writing no formal paper exists; the announcement is a machine-checkable certificate, which makes an exact experimental program both possible and necessary. This manuscript maps the findings of that program (repository: CAOS.RESEARCH, problem `algebraic-geometry/jacobian-conjecture`; experiments cited as EXP-*nnn*). Each claim is labeled: [MV] machine-verified exact identity, [D] derived analytically and certified on instances, [C] conjecture.

2 Exact validation and the full fiber (EXP-001)

Theorem 2.1 ([MV]). *For F as in (1): $\det JF = -2$ identically; the three announced points are pairwise distinct with the common image $(-\frac{1}{4}, 0, 0)$; and the fiber $F^{-1}(-\frac{1}{4}, 0, 0)$ consists of exactly those three points. In particular F is a non-injective Keller map and $JC(N)$ is false for every $N \geq 3$.*

The fiber computation is a lex Gröbner basis over $\mathbb{Q}$: the eliminant in z is $\frac{1}{8}(2z - 13)(4z + 1)$, and on the fiber $3x + 2y = 0$, $12y^2 = 4z + 1$. All coefficients of (1) are rational, so non-injectivity holds equally over $\mathbb{R}$.

3 The structure of the counterexample (EXP-002)

Theorem 3.1 ([MV]). *F is equivariant for the $\mathbb{G}_m$ -action with source weights $(1, -1, -2)$ and target weights $(-2, -1, 1)$. Writing $v = xy$, $t = x^2z$, $u = 1 + v$, the reduced data*

$$a_1 = x^2P, \quad b_1 = xQ, \quad c_1 = R/x = 2 - 3v - t$$

are polynomials in (v, t) ; each component of F is affine in z with coefficient vector $(u^3, 3xu^2, -x^3)$; and the generating identity $a_1 = \frac{ub_1}{2} - u^2 + \frac{u^3c_1}{2}$ holds.

Lemma 3.2 (Skew-product determinant lemma, [MV] with generic symbolic coefficients). *For any polynomials a_1, b_1, c_1 in (v, t) , the map $(a_1/x^2, b_1/x, xc_1)|_{v=xy, t=x^2z}$ satisfies*

$$\det JF \cdot c_1^2 = -J_2, \quad J_2 = \det \frac{\partial(V', T')}{\partial(v, t)}, \quad V' = b_1c_1, \quad T' = a_1c_1^2.$$

For F , $J_2 = 2c_1^2$ [MV]: the Keller property is a *two-variable* identity; the third dimension only carries weights. Fibers are governed by a single cubic: with $w = uc_1/2$ and $\Phi(w) = w^2 - w^3$,

$$\Phi(w) = \frac{1}{4} (wQR - PR^2) \quad \text{identically [MV]}, \quad (2)$$

so preimages of (A, B, C) correspond to roots of $4\Phi(w) - wBC + AC^2 = 0$; at the announced target this degenerates to $\Phi(w) = 0$ with roots $\{0, 0, 1\}$, matching the three preimages' w -values $(1, 0, 0)$ [MV]. Injectivity dies by *non-properness*: a Keller map is a local diffeomorphism, so sheets merge only at infinity; this is Pinchuk's loophole and the modern face of Kraus's gap.

4 The seed family: a general counterexample constructor (EXP-003/004)

The identification $c_1 = h(v) - t$ with $h = p/w$ is a degree-2 coincidence: an initial constructor built on it is *provably unsolvable* for $\deg p \geq 3$ (EXP-003, an exact unsolvability statement). The failure analysis yields the correct *potential form*.

Theorem 4.1 (Constructor v2, [D], certified on instances [MV]). *Let $p(w) = \sum_{i=1}^d p_i w^i$ satisfy*

$$p(0) = 0, \quad \int_0^1 p = 0, \quad p(1) \neq 0, \quad p'(1) \neq 2p(1),$$

let $k \neq 0$, $m = -kp(1)$, and let $q(v)$ be any polynomial with $q(0) = k$ and $q'(0) = k(p(1) - p'(1))/(p'(1) - 2p(1))$, with arbitrary higher-order tail. Set $\Phi(w) = \int_0^w p$, $s = c_1 = q(v) - t$, $w = us/k$, and define

$$V' = k^2 p(w) + ms, \quad T' = wV' - k^2 \Phi(w), \quad b_1 = V'/s, \quad a_1 = T'/s^2.$$

Then a_1, b_1 are polynomials in (v, t) , and $F_{p,k,q} = (a_1/x^2, b_1/x, xc_1)$ is a polynomial Keller map of $\mathbb{C}^3$ with

$$\det JF_{p,k,q} = -\frac{m^2}{k} = -kp(1)^2 \quad (\text{constant, for every admissible } (p, k, q)),$$

generic fiber degree $d + 1$, and the exact reconstruction inverse: from a root w of the fiber polynomial $k^2 \Phi(w) - (wBC - AC^2)$,

$$s = \frac{BC - k^2 p(w)}{m}, \quad u = \frac{kw}{s}, \quad x = \frac{C}{s}, \quad v = u - 1, \quad t = q(v) - s, \quad y = \frac{v}{x}, \quad z = \frac{t}{x^2}.$$

The announced map (1) is the instance $p = 2w - 3w^2$, $k = 2$, q affine; at $d = 2$ this shape is rigid up to the scale k [MV].

The three admissibility conditions are exactly the low-weight polynomiality constraints at the origin; the potential form makes $J_2 = m^2 s^2 / k$ automatic, so *the Keller property costs nothing*: this is the structural reason the conjecture is false in dimension ≥ 3 .

Theorem 4.2 (New explicit counterexamples, [MV]). *The following maps are non-injective polynomial Keller maps of $\mathbb{C}^3$, each certified by an engineered rational collision:*

Seed p	k	$\det$	Component degrees	Generic fiber degree
$w - 2w^3$	3	-3	(12, 11, 4)	4
$8w - 12w^2 + 4w^3 - 5w^4$	14	-350	(17, 16, 4)	5
$2w - 3w^2$, section tail v^2	2	-2	(8, 7, 5)	3

For the first: $(-\frac{1}{15}, 18, 5130)$ and $(\frac{1}{24}, -18, -10368)$ both map exactly to $(-54, -54, 1)$. Collisions are engineered by prescribing two rational roots $w_1 \neq w_2$ and solving the linear system $w_i \tilde{V} - \tilde{T} = k^2 \Phi(w_i)$ for the target invariants.

The section-tail instance changes the component-degree pattern at fixed fiber degree and determinant: the family is strictly larger than the seed-only family circulating informally. Every negative rational number is realized as a Keller determinant within the family.

Theorem 4.3 (Degree laws and family-internal minimality, [MV] for $d \leq 5$). *For affine sections q the component total degrees of $F_{p,k,q}$ are exactly $(5d - 3, 5d - 4, 4)$, $d = \deg p$ (verified $d = 2, \dots, 5$; the $d = 5$ instance, seed $w - 3w^5$, $k = 5$, $\det = -20$, degrees $(22, 21, 4)$, fiber degree 6, carries its own rational collision certificate). The scale k never changes degrees; section tails strictly increase them; degree-1 seeds are impossible (the integral condition forces $p = 0$), so the fiber-degree floor is 3. Consequently the minimal degree vector over the family is $(7, 6, 4)$, attained exactly at the announced map: the July 2026 counterexample is the smallest instance of its own family. Globally, only Wang’s degree-2 theorem constrains $\mathbb{C}^3$; the existence of a counterexample of total degree 3 to 6 remains open.*

Remark 4.4 (Characteristic p , [MV]). The $d = 3$ instance reduces mod $\ell = 13$ and $\ell = 101$ to explicit non-injective Keller maps over $\mathbb{F}_\ell$ of total degree $12 < \ell$ (the rational collision reduces to distinct points with a common image, e.g. mod 13: $(6, 5, 8)$ and $(6, 8, 6)$ both to $(11, 11, 1)$). Unlike the classical char- p example $x - x^p$ (degree $= p$), the degree sits strictly below the characteristic; expected from standard integer-model reductions, now explicitly certified.

5 The asymptotic variety: where sheets merge

Theorem 5.1 (Escape = multiple fiber root, [MV] on F and the $d = 3$ instance). *In the potential form, the fiber polynomial $W(w) = k^2 \Phi(w) - wBC + AC^2$ satisfies $W'(w) = k^2 p(w) - BC$ identically, and the reconstruction scale is $s = -W'(w)/m$. Hence a fiber root fails to reconstruct a finite preimage (the preimage escapes to infinity) precisely when it is a multiple root of W : over $C \neq 0$ the non-properness locus is the discriminant locus of W . For the announced F ,*

$$A(F) = \{C = 0\} \cup \{27A^2C^2 - 18ABC + 16A + B^3C - B^2 = 0\}.$$

Verified end to end: the target $(0, 1, 1)$ lies on the surface and its exact fiber is the single point $(2, -\frac{1}{2}, \frac{9}{8})$ (two sheets escaped); over generic $(A, B, 0)$ the fiber is the single flat-sheet point $(0, B, A - 4B^2)$; the collision target $(-16, -16, 1)$ lies off the surface with a full 3-point rational fiber. Collisions are generic; escapes are codimension 1. Kraus’s “ramification at infinity” is, for this family, the discriminant of a univariate polynomial.

6 The two-dimensional frontier: obstruction, rigidity, real census (EXP-005/006/010/011)

Theorem 6.1 ([MV] with generic symbolic f, g). *Let $\mathbb{G}_m$ act on $\mathbb{C}^2$ with weights $(1, -a)$, $a \geq 1$, and let $F = (x^{b_1}f(v), x^{b_2}g(v))$, $v = x^ay$, be an equivariant polynomial map. Then*

$$\det JF = x^{b_1+b_2+a-1} (b_1fg' - b_2f'g),$$

so the Keller condition forces $b_1 + b_2 = 1 - a$ and the single univariate identity $b_1fg' - b_2f'g = \text{const} \neq 0$.

Proposition 6.2 (One-invariant collapse). *If the reduced data (V', T') of a weighted skew-product depend on a single variable, then $dV' \wedge dT' = 0$, hence $J_2 \equiv 0$ and $\det JF \equiv 0$: no Keller map arises. Since a nontrivial $\mathbb{G}_m$ -action on $\mathbb{C}^2$ has one independent weight-0 invariant (against two in $\mathbb{C}^3$), the potential-form mechanism of Theorem 4.1 is structurally unavailable in dimension 2.*

Additionally [MV]: no coordinate slice of (1) (each variable frozen at rational test values, any output pair) is a 2D Keller map.

Theorem 6.3 (2D equivariant rigidity, [D], every mechanical step [MV] on sweeps). *For any $\mathbb{G}_m$ -action on $\mathbb{C}^2$ with weights $(w_1, -w_2)$, $w_1 \geq 1$, $w_2 \geq 0$, every equivariant polynomial Keller map is linear: (f_0x, g_0y) or (f_0y, g_0x) .*

Proof shape. Equivariant components are $P = x^{i_1}y^{j_1}f(v)$, $Q = x^{i_2}y^{j_2}g(v)$ with $v = x^{w_2}y^{w_1}$, and the determinant factors as

$$\det JF = x^{i_1+i_2-1} y^{j_1+j_2-1} [(i_1j_2 - i_2j_1)fg + v(\beta_1fg' - \beta_2f'g)], \quad \beta_r = w_1i_r - w_2j_r$$

(certified on 648 weight-times-exponent cases with generic symbolic f, g). Constancy forces $i_1 + i_2 = j_1 + j_2 = 1$, and the mixed shapes ($P = xyf$, $Q = g$ and its swap) admit no Keller solution (their bracket is divisible by v ; empty solves). The surviving shapes reduce to $fg + v(w_1fg' + w_2f'g) = c$ (or its swap), whose coefficient at the top bidegree $N = d_f + d_g$ is $(1 + w_1d_g + w_2d_f)f_{d_f}g_{d_g}$ with a strictly positive factor: the leading coefficients die and by descent f, g are constant (certified: empty constrained solves for all exact bidegrees ≤ 5 over eight weight pairs). $\square$

The independent enumeration EXP-006 (216 polynomial Keller instances over a bounded window, all 648 exactly-computed in-image fibers single-point, every instance linear [MV]) is the experimental companion of Theorem 6.3. Consequence: a two-variable counterexample, if one exists, must be genuinely non-equivariant: the symmetry class behind the entire three-dimensional family is completely closed in dimension 2.

Theorem 6.4 (Real fiber census of F , [MV]). *Over $\mathbb{R}$, the announced F restricts to a surjective, orientation-reversing ($\det = -2$), non-proper, non-injective real Keller map. For real targets off the discriminant wall the number of real preimages is 3 (region $D > 0$, which contains the collision targets) or 1 (region $D < 0$); two independent exact routes (Sturm isolation of the fiber cubic plus reconstruction, versus Gröbner elimination) agree at all samples, and no empty real fiber exists on a 36-target rational grid (an odd-degree argument makes this structural). In particular the constant-Jacobian real statement in dimension 3 falls with the same example; Pinchuk's planar counterexample (1994) concerned the non-constant case.*

The honest map of remaining routes to a hypothetical 2D counterexample: no symmetry at all; non-reductive ($\mathbb{G}_a$) structures, which face the same one-invariant collapse in naive form; or gradings at infinity in the Newton-polygon/Abhyankar sense. Conversely, a proof of $JC(2)$ in this program's terms is a properness statement: the reconstruction inverse of Theorem 4.1 shows exactly how asymptotic values are produced in dimension 3 (denominators $BC = k^2 p(w)$), and whether one-invariant geometry leaves room for such escape is, in our view, the sharpest transferable open question.

7 The planar machine: uniform closure, descent, and the primitive stratum

The Keller condition $\det J(P, Q) = 1$ is bilinear in the coefficient vectors of P and Q , hence linear in one side given the other. In the affine gauge (identity linear parts) this yields a machine per degree pair (m, n) : eliminate the large side (a lexicographic Groebner elimination produces the consistency ideal in the small side's coefficients), parametrize the consistency variety, complete linearly, and invert explicitly or fiber-test. The $(2, 3)$ consistency ideal is the single equation $(4A_0A_2 - A_1^2)^2 = 0$: the top form of P must be a perfect square, exactly the leading-form degeneracy; the $(2, 4)$ elimination returns the same ideal.

Theorem 7.1 (Uniform closure at min degree ≤ 2 ; machine-verified at the certified degrees). *Write $\ell = \alpha x + \beta y$, $\beta \neq 0$. In shear coordinates $(u_1, u_2) = (\ell, x)$ the Keller condition for $P = x + \ell^2$ is the linear first-order PDE $2u_1Q_{u_2} - Q_{u_1} = c$ with characteristic invariant P ; its complete polynomial solution space is $Q = \ell/\beta + H(P)$, H arbitrary. Consequently every planar Keller map with $\min(\deg P, \deg Q) \leq 2$ is invertible, with the closed inverse $\ell = \beta(v - H(u))$, $x = u - \ell^2$, $y = (\ell - \alpha x)/\beta$.*

Sufficiency is verified generically; completeness as exact kernel dimensions $\lfloor n/2 \rfloor + 1$ ($n = 3..6$, three rational (α, β) samples); the inverse by exact composition. The closure extends verbatim to any section polynomial: the whole quasi-triangular class $P \sim x + f(\ell)$ is invertible at every degree by the same formula with f in place of ℓ^2 . At bidegrees $(3, 3)$ the machine also forces the alignment: for $P = x + A_0x^2 + A_1xy + A_2y^2 + ay^3$ the consistency ideal contains A_0^2 and A_1^6 once a is inverted, so a cube-top cubic P admits completions only in quasi-triangular form.

The classical degree descent is, on this machine, an algorithm: while the leading forms are power-proportional, subtract cP^k from Q ; when the minimum degree reaches 2, finish with the closed inverse of Theorem 7.1; replay the elementary steps on the value side to assemble the inverse of the original map. The implementation explicitly inverted an entire deterministic library of composed Keller maps (0 to 4 steps each), verified by exact composition. Its failure signature is precise: a *primitive* pair, i.e. leading forms sharing a base form h with $\deg h \geq 2$ and no power relationship. $JC(2)$ is exactly the statement that no Keller pair is primitive.

First contact with that stratum, at the smallest primitive bidegrees $(4, 6)$ ($P_{\text{top}} = ah^2$, $Q_{\text{top}} = bh^3$, h quadratic): on a structured sampled slice ($h = xy$ and the rank sweep $x^2 + \gamma y^2$, 25 samples, $a \neq 0$, including a generic symbolic a), the completion window is empty: no Keller partner of degree ≤ 6 exists at all (the complete linear system is inconsistent in every sample), and one descent step extends the exclusion to degree ≤ 8 . Positive controls certify the scan is not vacuous. Verified against the primary sources, this replicates with a different instrument classically covered territory: Magnus proved invertibility for $\gcd = 1$ [?], and Appelgate–Onishi [?] with Nagata extended this to $\gcd$ prime (refined coverage: $\gcd \in \{1, 8\} \cup P \cup 2P$); the plane

conjecture is classically equivalent to the divisibility statement, which is exactly the primitive-stratum framing above. The genuinely open gcd values start at 9 and 12 (bidegrees like (18, 27) and (24, 36)): that composite-gcd ladder is the frontier. First contact is made there: for $h = x^4y^5$, the machine certifies that $P = x + ah^2$ admits NO Keller partner of degree ≤ 27 for any $ae0$ (a single certificate pairing with gcd $-144a^2$ over 403 completion unknowns), with the same emptiness across three structurally different degree-9 bases and a reach probe at (24, ≤ 36). Stated honestly: these bidegrees lie inside Moh's degree-100 verified range, so the contribution is the explicit per-pair certificate rather than the truth itself; the beyond-Moh rungs are the standing target.

Theorem 7.2 (The weight-class theorem; machine-verified at every step). *For $u \geq 2$, $v \geq 1$, $a \neq 0$, the polynomial $P = x + ax^uy^v$ is not a Keller component: no $Q \in \mathbb{C}[x, y]$, of any degree, satisfies $J(P, Q) \in \mathbb{C}^\times$.*

Proof sketch. Under the weights $(v, 1-u)$ the polynomial P is weighted-homogeneous, so $J(P, \cdot)$ shifts weight by a constant and the Keller equation decouples into independent weight classes. The constant 1 is reachable only from the class of y , spanned by the ray $g_s = x^{ks}y^{ms+1}$ with $k = (u-1)/d$, $m = v/d$, $d = \gcd(u-1, v)$; on that ray the operator is banded, $J(P, g_s) = (ms+1)e_s + aB_s e_{s+d}$ with B_s a positive integer and $e_t = x^{kt}y^{mt}$. For a polynomial Q the class component is a finite sum $\sum_{s \leq S} c_s g_s$: the e_0 row forces $c_0 \neq 0$, the band recursion keeps the chain nonzero, and the row beyond the last entry demands it vanish: a contradiction for every truncation S . $\square$

This is the closed form of every pure-slice window certificate above (all seven measured pairings are its chain products), and it certifies past every verified range at negligible cost (e.g. deg $P = 135$, beyond the degree-108 floor of the current literature). The proof is elementary once seen and may be folklore; we have not found it stated. Its scope is exactly the one-monomial perturbations; non-monomial slices have genuine multi-edge Newton polygons, and the multi-edge calculus (edge filtration, conjectured upper-triangularity) is the program's next instrument.

Theorem 7.3 (Lower-weight perturbations never rescue; machine-shadowed, one derived step). *Let $u \geq 2$, $v \geq 1$, $a \neq 0$, and let $R \in \mathbb{C}[x, y]$ be any polynomial whose monomials have degree ≥ 2 and $(v, 1-u)$ -weight strictly below v . Then $P = x + ax^uy^v + R$ is not a Keller component.*

Proof skeleton. The operator $L_{\text{top}} = J(x + ax^uy^v, \cdot)$ raises weight by $u-1$, strictly more than any perturbation term, so the top-graded equations are governed by L_{top} alone. Its kernel on the class of weight kv is exactly $\langle (x + ax^uy^v)^k \rangle$ (machine-identified; the naive injectivity claim fails precisely there, a refutation recorded in the experiment log). Absorbing these kernel components of Q as $H(P_{\text{top}})$ turns the perturbation into the sources $J(R, P^k)$ in the constant's equation, which Lemma 7.6 annihilates (they are images of $L = J(P, \cdot)$, and the certificate covector kills that image). The constant's equation then reduces to the pure chain, contradicted at every truncation. Certificates over $\mathbb{Q}(a, b, c)$ confirm the conclusion directly: with danger-weight tails present, the obstruction pairing remains a pure power of a . $\square$

This covers dense-Newton-polygon families of unbounded degree, far beyond every verified range, while remaining silent, as it must, on quasi-triangular components (their leading monomials violate $u \geq 2$, $v \geq 1$). The frontier moves to perturbations of weight $\geq v$, where the top edge itself deforms.

Theorem 7.4 (Every x -anchored edge falls; leading form unconditional). *Let $z = x^k y^m$ with $k, m \geq 1$ and $E = x\varphi(z)$ with $\deg \varphi = g \geq 1$. Then E is never the leading form of a Keller component of $\mathbb{C}^2$, and $P = E + R$, for any polynomial tail R of strictly lower weight, is never a Keller component, at any degree.*

Proof of the leading-form statement. On the y -class ray $g_s = x^{ks} y^{ms+1}$ the operator acts by multiplication: $J(E, g_s) = [(ms+1)\varphi(z) + kz\varphi'(z)]z^s$ (verified symbolically in the edge coefficients). The class equation therefore asks for a polynomial f with $mz\varphi f' + (\varphi + kz\varphi')f = 1$. If $\deg f = D$, the coefficient of z^{D+g} on the left is $(mD + kg + 1)\varphi_g c_D$, and every factor is nonzero; so $c_D = 0$, and by induction $f = 0$, contradicting the constant. The with-tails statement follows the kernel-absorption route of Theorem 7.3, with kernels now the powers of the full edge E (machine-identified). $\square$

Machine confirmations: full completion windows are empty for binomial and trinomial edges including perfect-square and perfect-cube φ (factorization does not rescue); the $\mathbb{Q}(a, b)$ certificate for the binomial edge is $60a^3$, free of the lower edge coefficient. Consequently no Keller component of $\mathbb{C}^2$ has a Newton top edge anchored at the linear vertex with interior direction: components must carry y -anchored (quasi-triangular type) tops, and the two-variable conjecture is exactly the statement that nothing else lives there. Classifying the y -anchored completions is the program's queued endgame frame.

Theorem 7.5 (The vertex dichotomy). *Let P be a Keller component of $\mathbb{C}^2$ with linear part x . If the monomial x is a vertex of the Newton polygon $N(P)$, then $P = x + f(y)$ for some univariate f .*

Proof. The two axis directions of the edge fan fall by one-line class arguments: for $P = x\varphi(y)$, weights $(1, 0)$ make the operator preserve each x -degree class, and the constant's class equation is $\varphi(y)h'(y) = 1$, impossible for a polynomial h once $\deg \varphi \geq 1$; for $P = x\varphi(x)$, $J(P, Q) = (x\varphi)'(x)Q_y$ forces a non-polynomial Q_y . Together with Theorem 7.4 (interior directions), every edge of $N(P)$ at the vertex $(1, 0)$ whose lattice points are all divisible by x is excluded; any support monomial $x^i y^j$ with $i \geq 1$, $(i, j) \neq (1, 0)$ would create such an edge as the first right-boundary edge at the vertex. The only admissible edges join $(1, 0)$ to pure- y vertices $(0, d)$, and such segments contain no interior lattice points. Hence the support of P is $\{x\} \cup \{y^j\}$. $\square$

The dichotomy's other side is real: $x + (x+y)^2$ and $x + (x+2y)^3$ are components in which x is swallowed inside the polygon. Every non-triangular component is of that kind, and a linear gauge rotation moves its dominating direction onto a coordinate, where the vertex analysis re-applies: the recorded strategy is the resulting induction, whose closure on the triangular family would be exactly the two-variable Jacobian conjecture. The machine's verification covers all three sides on structured samples (vertex- x mixed shapes inconsistent; triangular shapes consistent; swallowed shapes consistent).

Lemma 7.6 (Annihilation). *Let P be a polynomial, $L = J(P, \cdot)$, and let Λ be a covector annihilating the image of L on a space of polynomials containing $P^{k-1}m$. Then Λ annihilates the source $J(m, P^k)$.*

Proof. Since $J(P, P) = 0$, the Leibniz rule gives $J(P, P^{k-1}m) = P^{k-1}J(P, m)$, hence $J(m, P^k) = -k P^{k-1}J(P, m) = -k L(P^{k-1}m)$: the source is an image of L with the explicit preimage $P^{k-1}m$, and Λ kills it. $\square$

The certificate covectors used throughout are left-null on the completion matrix, whose columns are exactly $L(m)$ for the ansatz monomials, so they satisfy the hypothesis automatically. The only finite-dimensional subtlety is that the window must contain the preimage; truncated computations show spurious nonzero pairings for sources whose preimage exceeds the window, and the pairing vanishes precisely when $\deg(P^{k-1}m)$ fits, which reproduces the recorded artifact values exactly. With this lemma the tail steps of Theorems 7.3, 7.4 and 7.5 are unconditional.

The corner is diagonal; the residual core is the staircase

For a mixed corner $B = x^i y^j$ ($i, j \geq 1$) the corner operator is diagonal,

$$J(B^p, x^a y^b) = p(ib - ja) x^{ip+a-1} y^{jp+b-1},$$

with kernel exactly the powers of B , and no monomial makes $J(B^p, \cdot)$ carry a constant term. The mixed corner, long the suspected hard object, is therefore clean; what resists is the transport of the Keller constant along the staircase of Newton vertices joining the swallowed linear vertex to the corner. That staircase, the classical shape studied by Moh, is the residual open core, and the theorems above apply to it edge by edge.

8 Discussion and program

Why the conjecture is false. In the equivariant class, being Keller is a two-variable potential identity, satisfiable for free at any fiber degree ≥ 3 ; nothing in the determinant sees properness. **What is essential.** Two independent weight-0 invariants; the potential form; valuation conditions at the origin. **Consequence cascade (verified from primary sources, EXP-016).** The Mathieu conjecture is false for $SU(3)$ (Mathieu 1997: Mathieu for $SU(N)$ implies $JC(N)$; abelian case Duistermaat–van der Kallen 1998); the Gaussian moments conjecture is false (GMC implies JC, Derksen–van den Essen–Zhao 2017); Zhao’s vanishing conjecture is false (equivalent to JC); the Image conjecture is false in some dimension (implies the vanishing conjecture); and the full Dixmier conjecture is false (stable equivalence $JC(2n) \Leftrightarrow \text{Dixmier}(n)$, Belov-Kanel–Kontsevich and Tsuchimoto: $JC(4)$ false gives Dixmier(2) false; Dixmier(1) remains open). **Next experiments.** EXP-010 (2D widening: general weight pairs, higher degrees); the global-minimality search in degrees 3 to 6 (GPU-widened ansatz enumeration); the Bass–Connell–Wright push-through of F to explicit cubic form; primary-source verification of the consequence cascade.

The weighted landscape: uniqueness of the mechanism (EXP-012)

Generalizing to source weights $(1, -1, -m)$, the determinant lemma $\det JF = x^{\sigma+m} D$ and the pairing reductions $J_2 = \pm c_1^{r_a+r_b-1} D$ hold generically [MV], and the reduced Jacobian vanishes to order $m-1$ along the section. The survey outcome: at $m=1$ the reduced pair is itself a planar Keller map, so that class is a *bridge* to $JC(2)$ (93 scanned instances, all injective; a collision there would refute $JC(2)$); at $m=3$ the class $o = (-2, -2, 1)$ is *empty* (a one-line valuation proof, machine-certified) and $o = (-3, -1, 1)$ is rigid in scans (60 instances); at $m=4$ the potential-form family does *not* exist: with seed degree up to 5 the admissibility system has Gröbner basis [1] under $\mu \neq 0$ [MV]. Our design hypothesis (a parity law producing families at every even m) was thereby refuted by its own experiment; the finding is stronger: *within*

the scanned weighted landscape, the announced counterexample’s weight system is the unique collision-capable mechanism. Scope: sparse bounded scans, quantified in the experiment record; wider weight systems and higher-degree seeds are queued.

Reproducibility

Every theorem labeled [MV] is an exact sympy computation over $\mathbb{Q}$ with a persisted script and output in the repository (experiments EXP-001 through EXP-012 under `problems/algebraic-geometry/jacobi`; each experiment declares its hypothesis before the run and records an adversarial-validation route with its verdict.

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